

UNIVERSITY ADMISSION & ENROLLMENT SYSTEM



by

Aetisaam ul Hassan
BCY253051

A Project Report submitted to the
DEPARTMENT OF ELECTRICAL ENGINEERING
in partial fulfillment of the requirements for the degree of
BACHELORS OF SCIENCE IN CYBER SECURITY

Faculty of Engineering
Capital University of Science & Technology,
Islamabad
January, 2026

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It is declared that this is an original piece of my own work, except where otherwise acknowledged in text and references. This work has not been submitted in any form for another degree or diploma at any university or other institution for tertiary education and shall not be submitted by me in future for obtaining any degree from this or any other University or Institution.

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January, 2026

CERTIFICATE OF APPROVAL

It is certified that the project titled “**University Admission & Enrollment System**” carried out by **Aetisaam Ul Hassan**, Reg. No. **BCY253051** and **Aman Sajid** Reg. No. **BCY253084** under the supervision of **Muhammad Arslan Rafique**, Capital University of Science & Technology, Islamabad, is fully adequate, in scope and in quality, as a semester project for the degree of **BS CYBER SECURITY**.

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I would also like to acknowledge my **department and institution** for providing a supportive academic environment and necessary resources that made this project possible. Special thanks to my partner **Aman Sajid** for his cooperation, discussions, and hardwork throughout the project work.

Finally, I am grateful to my **family** for their constant encouragement and motivation, which helped me stay focused and committed to completing this project successfully.

ABSTRACT

University admissions usually involve multiple rules—minimum eligibility, program requirements, merit list placement, and fee or scholarship decisions. When this is done manually, it becomes slow and mistakes can happen, especially when many applicants are being checked at the same time. This project presents a simple **University Admission & Enrollment System** that automates this process using a console-based C++ program.

The system takes key applicant information such as **age, Matric and Intermediate percentages, entry test score, academic stream, locality status (hostel need), and family income**. After collecting inputs, the program first checks **basic eligibility**, then evaluates **program-specific selection criteria** (Computer Science, Engineering, Business/BBA, or Arts). It also calculates an overall **merit percentage**, assigns a **merit category**, and decides whether the applicant qualifies for **merit scholarship, need-based aid, or normal fee**. Finally, it displays hostel eligibility and a complete list of required documents for admission.

Overall, this project shows how ICT and programming logic can support real institutional decision-making by improving speed, consistency, and clarity. The program is easy to use, produces clear results, and can be extended later with features like a database, GUI, or online admission form integration.

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LIST OF ACRONYMS/ABBREVIATIONS

ICT	Information and Communication Technology
CS	Computer Science
BBA	Bachelor of Business Administration
GUI	Graphical User Interface
UI	User Interface
C++	C Plus Plus (Programming Language)
I/O	Input / Output
CNIC	Computerized National Identity Card
Merit %	Merit Percentage
Open Merit	General Merit-Based Admission
Self-Finance	Admission Based on Self-Funding
N/A	Not Applicable
Y/N	Yes / No
Std.	Standard
App.	Application
Dept.	Department

Chapter 1

INTRODUCTION

This chapter presents an overview of the **University Admission & Enrollment System** developed as an ICT project. It highlights the background, motivation, and core idea behind automating the admission decision process using programming logic. The chapter also outlines the project objectives, scope, and planned milestones to provide a clear understanding of the system's structure and purpose.

1.1 Overview

In today's competitive academic environment, universities receive a large number of applications every year. Managing these applications manually can be time-consuming, error-prone, and inconsistent, especially when multiple eligibility rules and merit criteria are involved. The use of Information and Communication Technology (ICT) helps automate such processes, making them faster, more accurate, and transparent.

This project focuses on developing a **University Admission & Enrollment System** using the C++ programming language. The system is designed to assist in evaluating student eligibility, suggesting suitable departments, determining merit categories, and providing admission-related decisions in a structured and logical manner.

1.2 Project Idea

The idea of this project is to create a **console-based admission decision system** that simulates the real admission process of a university. The program takes student data as input and processes it through multiple logical stages, including basic eligibility checks, program selection rules, merit calculation, and scholarship or fee assessment.

Instead of manual evaluation, the system automatically decides whether a student is eligible, which program they can apply for, their merit category, and whether they

qualify for hostel accommodation or financial aid. This makes the admission process more systematic and easy to understand.

1.3 Purpose of the Project

The main purpose of this project is to:

- Demonstrate the **practical application of C++ programming concepts**.
- Show how ICT can be used to **automate real-life decision-making systems**.
- Reduce human errors by applying **clear logical conditions** for admission decisions.
- Provide a simple and user-friendly tool that produces **clear and structured output**.

This project also helps students understand how programming logic can be used to solve administrative and institutional problems.

1.4 Project Objectives

The key objectives of this project are:

- To design a program that collects and processes student admission data.
- To apply eligibility rules based on **age, academic performance, and test scores**.
- To determine program selection according to **academic stream and merit criteria**.
- To calculate overall merit percentage and assign appropriate merit categories.
- To evaluate scholarship, financial aid, and hostel eligibility.
- To generate a complete checklist of required admission documents.

1.5 Applications of the Project

The University Admission & Enrollment System has practical applications in academic and administrative environments. The project demonstrates how ICT tools and programming logic can be used to automate admission-related decisions, improve efficiency, and reduce manual workload.

1.5.1 Automated Admission Decision Support

The system can be used to assist universities in automating the admission screening process. By evaluating student information such as age, academic results, and entry test scores, it quickly determines eligibility and program selection, ensuring consistency and reducing human error.

1.5.2 Merit, Scholarship, and Financial Aid Assessment

This project helps in calculating merit percentages and categorizing students into appropriate merit lists. It also supports decision-making for merit-based scholarships and need-based financial aid, enabling fair and transparent financial assistance policies.

1.5.3 Educational and Training Use

As an ICT learning project, the system is useful for educational and training purposes. It allows students to understand how real-world admission systems work while practicing programming concepts such as conditional statements, user input handling, and logical flow control.

1.6 Project Plan

The project plan outlines the systematic approach adopted for developing the University Admission & Enrollment System. It defines the sequence of activities, estimated time duration, and assigned responsibilities to ensure organized development. The plan is divided into two phases to effectively manage planning, implementation, testing, and documentation.

1.6.1 Project Milestones

Project milestones define the planned distribution of tasks, estimated time duration, and the responsible team members for each phase of the project. For effective management and smooth execution, the project has been divided into **two major parts (Part I and Part II)**. Each part focuses on specific activities ranging from planning and design to implementation and testing. This structured approach ensures timely completion and clear responsibility allocation among team members.

Table 1.1 Proposed Project Plan (Part 1)

S#	Tasks	Duration	Resource Person
1	Requirement Analysis and Problem Identification	01 Weeks	Aetisaam ul Hassan
2	Literature Review and Background Study	01 Weeks	Aetisaam ul Hassan, Aman Sajid
3	Flowchart, Presentation and Algorithm Design	01 Weeks	Aman Sajid
4	Input and Output Structure Planning	01 Week	Aetisaam ul Hassan
5	Project Proposal and Documentation	01 Weeks	Aman Sajid

Table 2.2 Proposed Project Plan (Part 2)

S#	Tasks	Duration	Resource Person
1	C++ Program Development	02 Weeks	Aetisaam ul Hassan
2	Implementation of Admission Logic	01 Weeks	Aetisaam ul Hassan, Aman Sajid
3	Testing and Debugging	01 Weeks	Aman Sajid
4	Output Verification and Evaluation	01 Week	Aetisaam ul Hassan
5	Final Report Writing and Formatting	01 Weeks	Aman Sajid

1.6.2 Project Timeline

The project timeline illustrates the planned schedule of activities for the **University Admission & Enrollment System**. The timeline is divided into Part I and Part II to clearly represent planning, design, implementation, and documentation phases. Microsoft Project–style Gantt charts are used to visually show task duration and progression, ensuring effective time management between 20 December and 10 January.

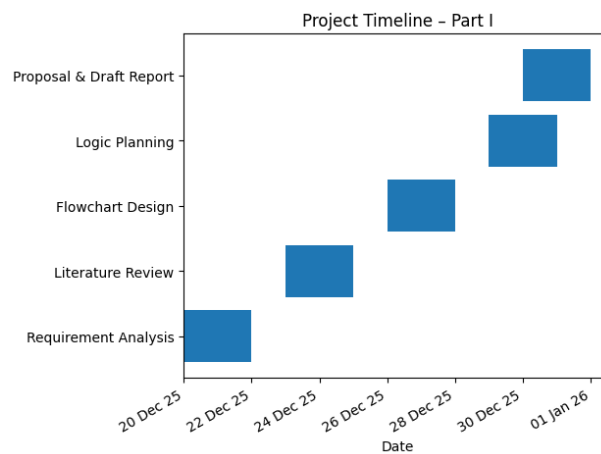


Figure 1.1 Project Timeline (Part 1)

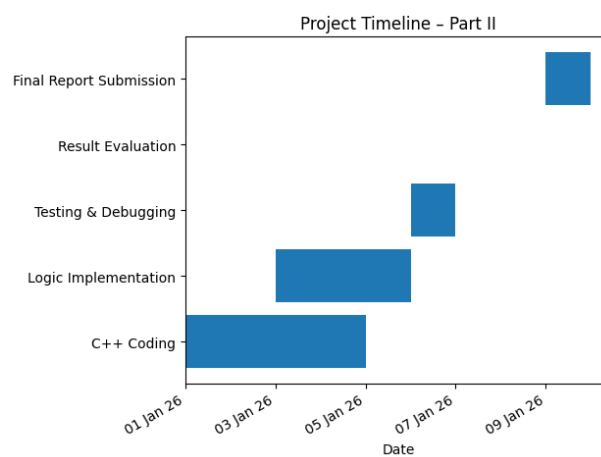


Figure 2.2 Project Timeline (Part 2)

1.7 Report Organization

This report is organized as follows:

- **Chapter 1** introduces the project, its background, objectives, and scope.
- **Chapter 2** presents the literature review and related concepts.
- **Chapter 3** explains the system design, methodology, and flowchart.
- **Chapter 4** discusses the tools and techniques used.
- **Chapter 5** evaluates the results and system performance.
- **Chapter 6** concludes the project and suggests future enhancements.

Chapter 2

LITERATURE REVIEW

2.1 Background Theory

University admission systems are an essential administrative component of higher education institutions. Traditionally, admission decisions have been carried out manually by reviewing application forms, academic records, and test results. This process often involves multiple eligibility rules, merit calculations, and category allocations, making it time-consuming and susceptible to human error. With the advancement of Information and Communication Technology (ICT), educational institutions have increasingly adopted automated systems to streamline admission processes and improve accuracy.

Admission decision systems generally rely on predefined criteria such as minimum age requirements, academic performance, entry test scores, and program-specific eligibility rules. These criteria are evaluated using logical decision-making mechanisms, which can be effectively implemented through programming. Automation helps ensure that all applicants are assessed fairly and consistently according to institutional policies.

From a theoretical perspective, this project is based on the concept of **Decision Support Systems (DSS)**. A DSS is a computer-based system that assists users in making decisions by processing input data using logical rules and conditions. In the context of university admissions, a DSS evaluates student data and produces outcomes such as eligibility status, merit category, scholarship decisions, and program allocation. This project applies these principles using a rule-based approach implemented in C++.

2.2 Related Technologies

The development of the **University Admission & Enrollment System** is supported by several closely related technologies that enable automation, logical decision-making, and reliable processing of student data. These technologies form the technical backbone of modern educational information systems and are widely used to manage administrative tasks efficiently. By leveraging such technologies, the project demonstrates how ICT tools can transform traditional manual admission procedures into structured, accurate, and transparent computerized processes.

The technologies discussed in this section are directly aligned with the objectives of the project and reflect real-world practices used in educational institutions. They provide both the practical implementation mechanism and the theoretical foundation required to automate admission-related decisions effectively.

2.1.1 Related Technology 1: C++ Programming Language

The **C++ programming language** is the core technology used in the implementation of this project. C++ is a powerful, high-performance language that supports structured and procedural programming, making it well-suited for developing logic-intensive applications such as admission decision systems. Its efficiency and flexibility allow developers to implement complex rules and conditions in a clear and controlled manner.

In this project, C++ is used to collect student information, evaluate eligibility criteria, and apply program-specific admission rules. Conditional statements are extensively utilized to check age limits, academic performance, entry test scores, and stream requirements. This enables the system to replicate real university admission policies accurately and consistently. Arithmetic operations are used to calculate merit percentages and determine admission categories, while formatted output ensures that results are presented clearly to the user.

Furthermore, C++ provides a simple yet effective console-based interface using standard input and output streams. This makes the system easy to understand and operate, especially in an academic environment where the focus is on learning programming logic rather than interface design. Due to its speed, reliability, and widespread academic use, C++ serves as an ideal technology for demonstrating ICT-based automation in this project.

2.1.2 Related Technology 2: Decision Support Systems (DSS)

The **University Admission & Enrollment System** is conceptually based on the principles of a **Decision Support System (DSS)**. A DSS is a computer-based system designed to assist in decision-making by analyzing data and applying predefined logical

rules. Such systems are commonly used in administrative and educational domains where multiple criteria must be evaluated objectively.

In the context of this project, the system acts as a rule-based DSS by processing student inputs and producing admission-related decisions such as eligibility status, program selection, merit category, scholarship or fee structure, and hostel eligibility. Each decision is derived from predefined admission rules, ensuring consistency and fairness for all applicants.

By adopting a DSS-based approach, the project highlights how ICT can support institutional decision-making processes. The system minimizes human error, reduces processing time, and improves transparency by clearly showing how decisions are made. Although implemented as a console-based application, the logical structure of the system can be easily extended to more advanced DSS implementations involving databases, graphical interfaces, or web-based admission portals.

2.3 Related Projects

Various academic and software projects have been developed to automate admission and enrollment processes. Some projects focus on fully **web-based admission portals** that allow students to apply online, upload documents, and track application status. These systems usually integrate databases, authentication mechanisms, and graphical user interfaces.

Other projects emphasize **rule-based admission systems**, where eligibility and selection are determined using predefined criteria. Such systems are often implemented using programming languages like C++ or Java and are suitable for demonstrating decision-making logic in educational environments.

Several student projects documented in academic repositories show that console-based admission systems are effective learning tools. These projects typically include features such as eligibility checking, merit calculation, and program recommendation. However, many of them are limited to basic selection logic and do not consider additional aspects like scholarship decisions, hostel eligibility, or document verification.

Compared to these works, the current project provides a more **comprehensive decision** framework by combining eligibility checks, program selection, merit categorization, financial aid assessment, and hostel eligibility in a single system.

2.4 Limitations and Bottlenecks of the Existing Work

Despite the availability of automated admission systems, several limitations exist in current implementations. Many systems are complex and require extensive infrastructure, making them difficult to deploy in smaller institutions or academic settings. Some systems rely heavily on databases and web technologies, which may not be suitable for introductory **ICT** learning environments.

Additionally, some existing systems lack transparency, as applicants may not fully understand how decisions are made. Others focus only on selection status and ignore related aspects such as merit categorization or financial assistance. Manual systems, which are still used in some institutions, are prone to errors, delays, and inconsistencies.

These limitations highlight the need for simplified, transparent, and logic-based admission systems that clearly demonstrate how decisions are made. Such systems are particularly valuable for educational purposes and for institutions seeking basic automation solutions.

2.5 Problem Statement

University admission processes involve multiple criteria and decision rules that must be applied consistently to all applicants. Manual handling of these rules increases the risk of errors and delays, especially when dealing with a large number of applications. Students may also face uncertainty due to a lack of clarity regarding eligibility, merit status, and financial assistance options.

The problem addressed in this project is the absence of a simple, transparent, and automated system that can evaluate student admission data and generate clear admission decisions using logical rules. There is a need for an **ICT**-based solution that

demonstrates how admission processes can be automated effectively using basic programming concepts.

2.6 Summary

This chapter reviewed the theoretical background and existing work related to automated university admission systems. It explained how traditional manual admission processes are time-consuming and error-prone, and how ICT-based solutions improve accuracy, consistency, and transparency. The chapter discussed the role of decision-making logic and highlighted Decision Support Systems (DSS) as the theoretical foundation for automating admission decisions.

Related technologies such as the C++ programming language and DSS were examined to show how they support rule-based evaluation of student data. A review of related projects demonstrated that while various admission systems exist, many lack comprehensive decision coverage or transparency. The limitations of existing systems highlighted the need for a simple, logic-driven, and educationally suitable solution.

Based on this analysis, the chapter justified the development of the proposed **University Admission & Enrollment System**, which aims to automate admission decisions using clear rules and programming logic. The next chapter focuses on the design and implementation of the proposed system.

Chapter 3

PROJECT DESIGN AND IMPLEMENTATION

This chapter describes the complete design and implementation process of the **University Admission & Enrollment System**. At this stage of the report, the objectives, background theory, and related technologies have already been discussed. This chapter focuses on how the system was designed step by step, the methodologies used, and the techniques applied to achieve the desired results. It also discusses alternative approaches considered during development and justifies the final design choices.

3.1 Proposed Design Methodology

The proposed design methodology follows a structured and stepwise approach to ensure clarity, correctness, and ease of implementation. The methodology begins with collecting user input, followed by eligibility evaluation, program selection, merit calculation, and final decision generation.

The system was designed as a console-based decision support application. Instead of using a database or web interface, a rule-based logic approach was selected to keep the system simple, transparent, and suitable for an academic ICT project. Each stage of the admission process is implemented using conditional statements that reflect real-world university admission policies.

The overall working of the system can be divided into the following main blocks:

- User Input Module
- Eligibility Evaluation Module

- Program Selection Module
- Merit Calculation Module
- Scholarship and Hostel Decision Module
- Output and Result Display Module

These blocks interact sequentially to produce a complete admission decision.

Figure 3.1: Block Diagram of the Proposed System Design

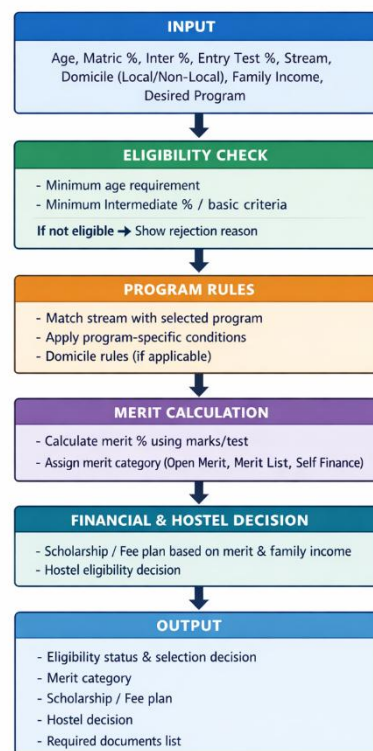


Figure 3.1: Block diagram of system design

3.2 Analysis Procedure

During the analysis phase, multiple approaches were considered for implementing the admission system. One option was to use a database-driven system where student records could be stored and processed. Another option was to design a graphical user interface (GUI) application. However, these approaches were found to be unnecessarily complex for the scope of this project and less suitable for demonstrating core ICT concepts.

The selected approach uses rule-based decision logic implemented in C++. This method was chosen because:

- It allows precise control over admission rules
- It is easy to debug and verify
- It clearly demonstrates decision-making using conditional statements
- It aligns with introductory ICT and programming learning outcomes

Alternative techniques such as file handling and database storage were explored conceptually but not implemented, as they did not significantly improve the learning objectives or results for this project. The console-based approach was therefore finalized as the most effective solution.

3.3 Design of the Project Software/Algorithm

The software design focuses on creating a clear logical flow that mirrors the real admission process of a university. The algorithm begins by collecting user input such as age, academic percentages, entry test score, academic stream, income level, and desired program.

The system then applies:

- **Basic eligibility checks** (minimum age and intermediate percentage)
- **Program-specific rules** (Computer Science, Engineering, Business, Arts)
- **Merit percentage calculation**
- **Merit category assignment**
- **Scholarship or fee decision**
- **Hostel eligibility determination**

A flowchart was created to visualize this logic and ensure that no decision step was missed.

3.4 Implementation Procedure

The implementation was carried out using the **C++ programming language**. The program starts by displaying a structured menu and prompts the user to enter required information. Input validation is applied to ensure meaningful and acceptable values.

The system then processes the input through multiple conditional blocks. Each block corresponds to a specific admission rule. Once all checks are completed, the system generates a detailed output displaying:

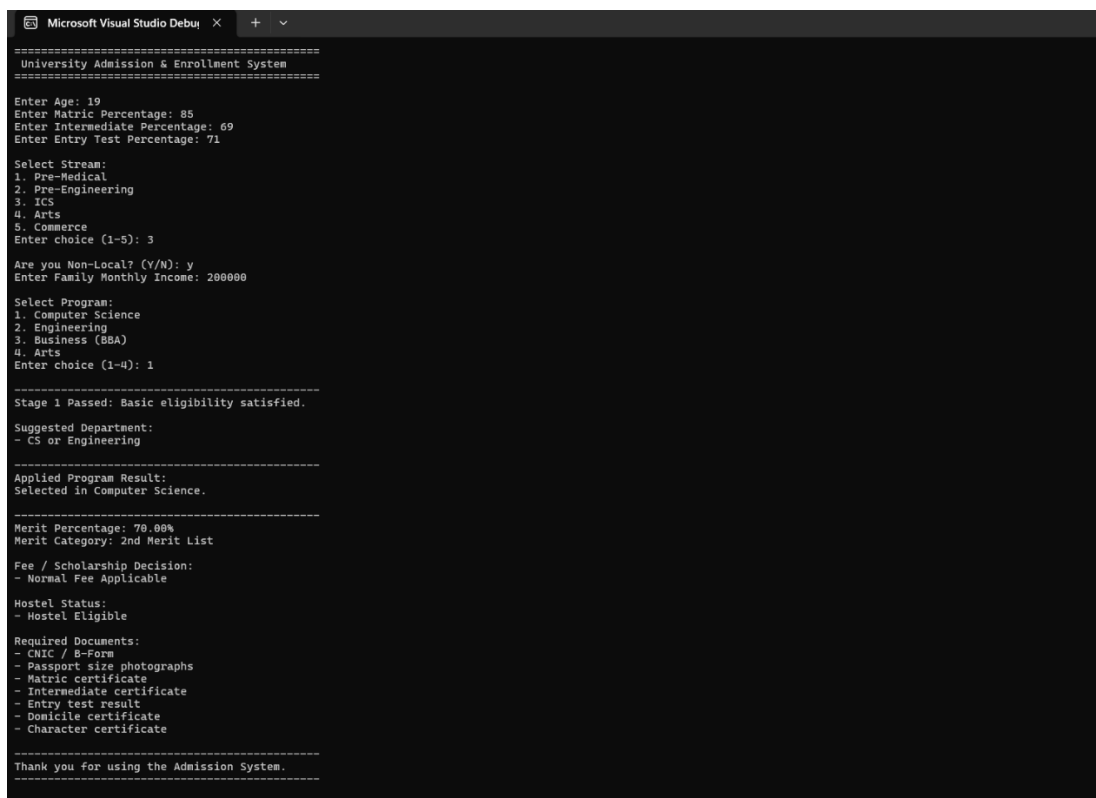
- Eligibility status
- Selected or non-selected program
- Merit percentage
- Merit category
- Scholarship or fee status
- Hostel eligibility
- Required document checklist

The implementation was tested using multiple sample inputs to verify correctness and consistency.

3.5 Details of Final Working Prototype

The final working prototype is a fully functional console-based application. The output screen displays all admission decisions in a clear and readable format. The reference output image *fig 3.5* demonstrates a successful execution of the system where user inputs are processed and results are generated correctly.

The prototype fulfills all project objectives and demonstrates a complete admission decision workflow from input to final output.



```
Microsoft Visual Studio Debug
=====
University Admission & Enrollment System
=====
Enter Age: 19
Enter Matric Percentage: 85
Enter Intermediate Percentage: 69
Enter Entry Test Percentage: 71
Select Stream:
1. Pre-Medical
2. Pre-Engineering
3. ICS
4. Arts
5. Commerce
Enter choice (1-5): 3
Are you Non-Local? (Y/N): y
Enter Family Monthly Income: 200000
Select Program:
1. Computer Science
2. Engineering
3. Business (BBA)
4. Arts
Enter choice (1-4): 1
=====
Stage 1 Passed: Basic eligibility satisfied.
Suggested Department:
- CS or Engineering
=====
Applied Program Result:
Selected in Computer Science.
=====
Merit Percentage: 70.00%
Merit Category: 2nd Merit List
Fee / Scholarship Decision:
- Normal Fee Applicable
Hostel Status:
- Hostel Eligible
Required Documents:
- CNIC / B-Form
- Passport size photographs
- Matric certificate
- Intermediate certificate
- Entry test result
- Domicile certificate
- Character certificate
=====
Thank you for using the Admission System.
=====
```

Figure 3.5: Final output

3.6 Summary

This chapter presented the complete design and implementation process of the **University Admission & Enrollment System**. It explained the adopted methodology, analyzed alternative approaches, and justified the selection of a rule-based software

design. The chapter also detailed the software architecture, algorithm design, and implementation procedure.

Through structured logic, flowchart-based planning, and systematic testing, the project successfully achieved its objectives. The system provides a clear example of how ICT tools and programming techniques can be applied to automate real-world administrative processes. The next chapter discusses the tools and techniques used during the development of this project.

Chapter 4

TOOLS AND TECHNIQUES

This chapter describes the tools and techniques used in the development of the **University Admission & Enrollment System**. It includes details of the hardware and software resources that supported the design, implementation, testing, and documentation of the project. The selected tools were chosen to ensure simplicity, efficiency, and alignment with the learning objectives of an ICT-based academic project.

4.1 Hardware Tools

The project is primarily **software-based** and does not require specialized hardware components. However, basic computing hardware was essential for development, execution, and testing of the system.

4.2 Software and Simulation Tools

Several software tools were used during different stages of the project, including development, testing, documentation, and design visualization.

4.2.1 C++ Programming Language

The **C++ programming language** was used to implement the core logic of the admission system. It enabled structured programming, conditional decision-making, and efficient handling of user input and output. C++ was selected due to its performance efficiency, widespread academic use, and suitability for implementing rule-based systems.

4.2.2 Integrated Development Environment (IDE)

An Integrated Development Environment such as **Dev-C++ / Code::Blocks / Visual Studio Code** was used for writing, compiling, and debugging the program. The IDE provided features like syntax highlighting, error detection, and debugging support, which improved development efficiency and code accuracy.

4.2.3 Flowchart and Diagram Design Tools

Flowcharting tools such as **Microsoft Visio** / **Draw.io** / **Lucidchart** were used to design the system flowchart and block diagrams. These diagrams helped visualize the admission decision process and ensured that all logical steps were properly planned before implementation.

4.2.4 Documentation Tools

Microsoft Word was used for preparing the project proposal and final report. It supported professional formatting, table creation, figure insertion, and consistent document styling according to academic standards.

4.3 Summary

This chapter presented the hardware and software tools used in the development of the University Admission & Enrollment System. Basic computing hardware enabled program execution and testing, while software tools such as C++, an IDE, flowcharting applications, and documentation tools supported development and presentation. The selected tools ensured efficient implementation, clear visualization, and professional documentation of the project. Together, these tools played a vital role in successfully achieving the project objectives.

Chapter 5

PROJECT RESULTS AND EVALUATION

This chapter presents the results obtained from the implementation of the **University Admission & Enrollment System** and evaluates the performance of the system based on different input scenarios. The chapter demonstrates how the system behaves under various conditions and assesses whether the project objectives have been successfully achieved.

5.1 System Output and Results

After successful implementation, the system produces clear and structured outputs based on the user's input. When the program is executed, the user is prompted to enter personal and academic information such as age, academic percentages, entry test score, academic stream, income level, and desired program.

Based on these inputs, the system displays:

- Basic eligibility status
- Suggested department
- Program selection result
- Merit percentage
- Merit category
- Scholarship or fee decision
- Hostel eligibility status
- Required document checklist

The output is displayed in a well-organized manner, making it easy for the user to understand the admission decision. The reference output image *fig 5.1* demonstrates a successful execution where all decision stages are processed correctly.

```
Microsoft Visual Studio Debug Console
=====
University Admission & Enrollment System
=====

Enter Age: 19
Enter Matric Percentage: 85
Enter Intermediate Percentage: 69
Enter Entry Test Percentage: 71

Select Stream:
1. Pre-Medical
2. Pre-Engineering
3. ICS
4. Arts
5. Commerce
Enter choice (1-5): 3

Are you Non-Local? (Y/N): y
Enter Family Monthly Income: 200000

Select Program:
1. Computer Science
2. Engineering
3. Business (BBA)
4. Arts
Enter choice (1-4): 1

-----
Stage 1 Passed: Basic eligibility satisfied.
Suggested Department:
- CS or Engineering

-----
Applied Program Result:
Selected in Computer Science.

-----
Merit Percentage: 70.00%
Merit Category: 2nd Merit List

Fee / Scholarship Decision:
- Normal Fee Applicable

Hostel Status:
- Hostel Eligible

Required Documents:
- CNIC / B-form
- Passport size photographs
- Matric certificate
- Intermediate certificate
- Entry test result
- Domicile certificate
- Character certificate

-----
Thank you for using the Admission System.
=====
```

Figure 5.1: Output

5.2 Test Case Analysis

To evaluate the correctness and reliability of the system, multiple test cases were applied by changing input values such as academic scores, age, and income level. These test cases helped verify whether the system responds accurately under different scenarios.

Test Case 1: High Academic Performance

- High intermediate and test scores
- Result: Selection on open merit with merit scholarship approval

Test Case 2: Average Academic Performance

- Moderate academic scores
- Result: Selection in second merit list or self-finance category

Test Case 3: Low Academic Performance

- Academic scores below minimum criteria
- Result: Not selected due to failure to meet eligibility requirements

Test Case 4: Non-Local Student

- Non-local status selected
- Result: Hostel eligibility confirmed based on merit

These test cases confirm that the system behaves as expected for different categories of applicants.

5.3 Performance Evaluation

The performance of the system was evaluated based on accuracy, efficiency, and usability. The program executes quickly, even when multiple decision rules are applied, due to the efficient nature of C++.

The rule-based logic ensures consistent and unbiased results for all users. Since the system is console-based, it is easy to operate and does not require advanced computing resources. The clarity of output improves user understanding of admission decisions.

5.5 Limitations Observed During Evaluation

Although the system performs well within its scope, certain limitations were observed during evaluation:

- The system does not store applicant data for future use
- No graphical user interface is provided
- Input validation is limited to basic checks

These limitations do not affect the core functionality but highlight areas for improvement in future versions.

4.3 Summary

This chapter analyzed the results produced by the University Admission & Enrollment System and evaluated its performance using different test scenarios. The system demonstrated accurate decision-making, reliable performance, and ease of use. The evaluation confirms that the project objectives were met successfully. The next chapter concludes the project and discusses possible future enhancements.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The primary outcome of this project is the successful development of a **University Admission & Enrollment System** that demonstrates how admission decisions can be automated using ICT and programming logic. The strongest conclusion drawn from this work is that even a simple, rule-based system can significantly improve clarity, consistency, and fairness in decision-making when compared to manual admission processes.

By translating real university admission policies into logical conditions, the system effectively evaluates eligibility, program suitability, merit category, financial assistance, and hostel requirements. The project confirms that structured programming and decision support concepts can be applied to solve practical administrative problems in an efficient and transparent manner. If discussed months later, the most important takeaway would be that **ICT tools, when applied thoughtfully, can simplify complex institutional processes without requiring advanced infrastructure.**

This work also provided valuable insight into how decision rules must be carefully designed to avoid ambiguity and ensure fairness. Through testing and evaluation, it was observed that clearly defined criteria lead to predictable and reliable outcomes. Overall, the project achieved its objectives and demonstrated the practical role of ICT in educational administration.

6.2 Future Work and Recommendations

Although the current system fulfills its intended purpose, several improvements can be made to enhance its functionality and real-world applicability. Future work could include the integration of a **database** to store and manage student records, allowing retrieval and modification of applicant data. This would make the system more suitable for large-scale institutional use.

Another possible enhancement is the development of a **graphical or web-based interface** to improve user interaction and accessibility. Implementing stronger input validation and error handling would further increase system reliability. Additionally, features such as online application submission, document upload, and automated merit list generation could be incorporated.

For students or developers who wish to continue this work, it is recommended to extend the system using modern ICT tools while preserving the clear decision logic established in this project. With these enhancements, the system can evolve from an academic prototype into a more comprehensive admission management solution.

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APPENDICES

Appendix – A

Code A

```
1  ✓ #include <iostream>
2  [ #include <iomanip>
3    using namespace std;
4
5  ✓ int main() {
6
7      int age;
8      double matric, inter, test;
9      int stream, program;
10     char nonLocal;
11     double income;
12
13     cout << "=====\\n";
14     cout << " University Admission & Enrollment System \\n";
15     cout << "=====\\n\\n";
16
17     // ----- INPUT -----
18     cout << "Enter Age: ";
19     cin >> age;
20
21     cout << "Enter Matric Percentage: ";
22     cin >> matric;
23
24     cout << "Enter Intermediate Percentage: ";
25     cin >> inter;
26
27     cout << "Enter Entry Test Percentage: ";
28     cin >> test;
29
30     cout << "\\nSelect Stream:\\n";
31     cout << "1. Pre-Medical\\n";
32     cout << "2. Pre-Engineering\\n";
33     cout << "3. ICS\\n";
```

Figure 3: Code A

Code B

```
34     cout << "4. Arts\\n";
35     cout << "5. Commerce\\n";
36     cout << "Enter choice (1-5): ";
37     cin >> stream;
38
39     cout << "\\nAre you Non-Local? (Y/N): ";
40     cin >> nonLocal;
41
42     cout << "Enter Family Monthly Income: ";
43     cin >> income;
44
45     cout << "\\nSelect Program:\\n";
46     cout << "1. Computer Science\\n";
47     cout << "2. Engineering\\n";
48     cout << "3. Business (BBA)\\n";
49     cout << "4. Arts\\n";
50     cout << "Enter choice (1-4): ";
51     cin >> program;
52
53     cout << "\\n-----\\n";
54
55     // ----- STAGE 1: BASIC ELIGIBILITY -----
56     ✓ if (age < 17) {
57         cout << "Not Eligible: Age must be 17 or above.\\n";
58         return 0;
59     }
60
61     ✓ if (inter < 50) {
62         cout << "Not Eligible: Intermediate must be at least 50%.\\n";
63         return 0;
64     }
65
66     cout << "Stage 1 Passed: Basic eligibility satisfied.\\n\\n";
```

Figure 4:Code B

Code C

```
67 // ----- DEPARTMENT SUGGESTION -----
68 cout << "Suggested Department:\n";
69 if (stream == 2 || stream == 3) {
70     cout << "- CS or Engineering\n";
71 }
72 else if (stream == 5) {
73     cout << "- Business (BBA)\n";
74 }
75 else if (stream == 4) {
76     cout << "- Arts / Social Sciences\n";
77 }
78 else {
79     cout << "- Arts or Business\n";
80 }
81
82 cout << "\n-----\n";
83
84 // ----- STAGE 2: PROGRAM SELECTION -----
85 cout << "Applied Program Result:\n";
86
87 if (program == 1) { // CS
88     if (inter >= 60 && test >= 55) {
89         cout << "Selected in Computer Science.\n";
90     }
91     else {
92         cout << "Not Selected in Computer Science.\n";
93         cout << "Required: Inter >= 60% and Test >= 55%\n";
94     }
95 }
96
97 else if (program == 2) { // Engineering
98     if (age > 23) {
```

Figure 5:Code C

Code D

```
100     cout << "Not Selected: Age must be 23 or below for Engineering.\n";
101 }
102 else if (stream != 2) {
103     cout << "Not Selected: Only Pre-Engineering students allowed.\n";
104 }
105 else if (inter >= 65 && test >= 60) {
106     cout << "Selected in Engineering.\n";
107 }
108 else {
109     cout << "Not Selected in Engineering.\n";
110     cout << "Required: Inter >= 65% and Test >= 60%\n";
111 }
112
113 else if (program == 3) { // BBA
114     if (inter >= 55 && test >= 50) {
115         cout << "Selected in Business (BBA).\n";
116     }
117     else {
118         cout << "Not Selected in Business.\n";
119         cout << "Required: Inter >= 55% and Test >= 50%\n";
120     }
121 }
122
123 else if (program == 4) { // Arts
124     if (inter >= 45) {
125         cout << "Selected in Arts.\n";
126     }
127     else {
128         cout << "Not Selected in Arts.\n";
129         cout << "Required: Inter >= 45%\n";
130     }
131 }
132 }
```

Figure 6: Code D

Code E

```
133     }
134     else {
135         cout << "Invalid program choice.\n";
136         return 0;
137     }
138
139     cout << "\n-----\n";
140
141     // ----- MERIT CATEGORY -----
142     double merit = (inter + test) / 2;
143     cout << fixed << setprecision(2);
144     cout << "Merit Percentage: " << merit << "%\n";
145
146     if (merit >= 80) {
147         cout << "Merit Category: Open Merit\n";
148     }
149     else if (merit >= 70) {
150         cout << "Merit Category: 2nd Merit List\n";
151     }
152     else if (merit >= 60) {
153         cout << "Merit Category: Self Finance\n";
154     }
155     else {
156         cout << "Merit Category: Not Selected\n";
157     }
158
159     // ----- SCHOLARSHIP / AID -----
160     cout << "\nFee / Scholarship Decision:\n";
161     if (inter >= 85 && test >= 70) {
162         cout << "- Merit Scholarship Approved\n";
163     }
164     else if (income <= 80000 && merit >= 70) {
165         cout << "- Need Based Aid Approved\n";
```

Figure 7:Code E

Code F

```
166     }
167     else {
168         cout << "- Normal Fee Applicable\n";
169     }
170
171     // ----- HOSTEL -----
172     cout << "\nHostel Status:\n";
173     if (nonLocal == 'Y' || nonLocal == 'y') {
174         cout << "- Hostel Eligible\n";
175         if (merit >= 80) {
176             cout << "- Hostel Priority: High\n";
177         }
178     }
179     else {
180         cout << "- Hostel Not Eligible\n";
181     }
182
183     // ----- DOCUMENT CHECKLIST -----
184     cout << "\nRequired Documents:\n";
185     cout << "- CNIC / B-Form\n";
186     cout << "- Passport size photographs\n";
187     cout << "- Matric certificate\n";
188     cout << "- Intermediate certificate\n";
189     cout << "- Entry test result\n";
190     cout << "- Domicile certificate\n";
191     cout << "- Character certificate\n";
192
193     cout << "\n-----\n";
194     cout << "Thank you for using the Admission System.\n";
195     cout << "-----\n";
196
197     return 0;
198 }
```

Figure 8:Code F

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Electrical and Computer Engineering Department

LAB PROJECT ASSESSMENT

1	Project Title	UNIVERSITY ADMISSIONS & ENROLLMENT SYSTEM		
2	Lab	CYG1611- Applications of Information and Communication Technologies Lab	Semester 01	Fall 2025
3	Student Name & Registration No.	Student 1	Student 2	Student 3
		AETISAAM UL HASSAN BCY253051	AMAN SAJID BCY253084	--
4	Instructor Name & Signature	Mr. Muhammad Arslan Rafique		

Project Demonstration

Assessment Criteria	Very Poor 0-1	Poor 2	Satisfactory 3	Good 4	Excellent 5	Score Student 1	Score Student 2	Score Student 3
Design Evaluation and Testing	No or very poor design prototype and demonstration.	Design prototype is not working and no testing of design has been done	Design prototype is partially functional and little testing of design has been done.	Design prototype is functional and some testing of design has been done.	Design prototype is fully functional and design has been exhaustively tested.			
Usage of software tools in design and evaluation	No or very poor software tools usage in project design and results evaluation	Insignificant evidence of software tools usage in project design and results evaluation	Little evidence of ability to select appropriate software tools, in project design and results evaluation	Some evidence of skills to use software tools in project design and results evaluation	Clear evidence of skills to use software tools in project design and results evaluation			

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Assessment Criteria	Very Poor 0-1	Poor 2	Satisfactory 3	Good 4	Excellent 5	Score Student 1	Score Student 2	Score Student 3
Knowledge of Project Implementation details (Q/A)	No or very poor knowledge of implementation and design process.	Poor knowledge of implementation and design with wrong/no answers	Satisfactory knowledge of implementation, vague answers	Adequate knowledge of project implementation with majority of correct answers	Exceptional knowledge of implementation and overall design with clear and spontaneous answers.			

Project Report

Assessment Criteria	Very Poor 0-1	Poor 2	Satisfactory 3	Good 4	Excellent 5	Score Student 1	Score Student 2	Score Student 3
Literature Survey, Problem Analysis and Design Procedure	No or very poor literature survey done. No problem analysis performed. No worthwhile design procedure exists.	Insufficient literature survey. Problem analysis part is skipped or does not contribute to creating an effective design. Does not follow any design procedure.	Partial literature survey. Problem Analyses performed is haphazard and design parameter selection is spontaneous. Little use of design procedure.	Adequate literature survey. Problem analysis performed correctly. Project demonstrates some use of design process.	Clear and complete literature survey, effective problem analyses is performed to choose design parameters. Project demonstrates effective use of design process.			
Language, Grammar and References	A lot of spelling and grammatical mistakes with poor English. The list of references is clearly inadequate. Table of content missing.	Frequent spellings and grammatical errors. The list of references should be expanded.	Occasional spellings and grammatical errors. The list of references appears reasonable but citation does not follow standard format.	Very few spellings and grammatical errors. Organization is good. The list of references appears reasonable and citation follow standard format.	Almost no spelling or grammatical mistake. Excellent organization. A comprehensive list of references is cited using the standard format.			